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Predicting the NHL playoffs with PageRank

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Abstract: Applying Google's PageRank model to sports is a popular concept in contemporary sports ranking. However, there is limited evidence that rankings generated with PageRank models do well at predicting the winners of playoffs series. In this paper, we use a PageRank model to predict the outcomes of the 2008–2016 NHL playoffs. Unlike previous studies that use a uniform personalization vector, we incorporate Corsi statistics into a personalization vector, use a nine-fold cross validation to identify tuning parameters, and evaluate the prediction accuracy of the tuned model. We found our ratings had a 70% accuracy for predicting the outcome of playoff series, outperforming the Colley, Massey, Bradley-Terry, Maher, and Generalized Markov models by 5%. The implication of our results is that fitting parameter values and adding a personalization vector can lead to improved performance when using PageRank models.

Keywords: Corsi statistics; Markov chains; National Hockey League (NHL).

1 Introduction

Predicting outcomes in the playoffs of any sport is challenging, but especially in hockey, because the conditions of play and possession constantly change. Teams quickly transition between offense and defense, players rotate during line changes, and powerplays create intervals of time where the offense has a player advantage.

Dominant regular season performances also tend to be less pronounced in the NHL, and playoff series tend to be drawn out. If we focus on the subset of teams that made the NHL playoffs over the seasons from 2008 to 2016, we find that the top and bottom ranked playoff teams had average winning percentages of 66% and 50% respectively in the regular season. In contrast, the top and bottom ranked playoff teams in the NFL had average winning percentages of 87% and 58% respectively. During the same time period, 34% of NHL playoff series went to game seven, 33% went to game six, 21% went to game five, and only 12% were decided in the minimum of four games. Hence, the small difference between the winning percentages of the top and bottom ranked teams and the relatively large proportion of seven-game series suggest the NHL playoffs generally feature more evenly matched teams.

As a result, the NHL playoffs can be incredibly difficult to predict, and most ranking systems struggle to achieve consistent results from year to year. The Colley and Massey models are two of the most well-known approaches for ranking teams and were used in the BCS college football rankings for over a decade until 2013 when the BCS formed a 13-member committee to determine rankings. The Colley model relies solely on the number of wins and losses and generates ratings by solving a system of linear equations (Colley 2002). This model is fairly straight forward and accounts for strength of schedule when determining ratings, but it does not consider margin of victory. In contrast, the Massey model develops a system of linear equations based on margin of victory and generates ratings using a least squares solution technique (Massey 1997).

Earlier research into pairwise comparisons produced a probabilistic model known as the Bradley-Terry model (Bradley and Terry 1952). This model has proven useful for calculating ratings based on observed comparisons which can be represented in a weighted, directed graph. The ratings, and consequential rank of a team, developed for each node can then be used to determine which team is more likely to win (Hunter 2004). This model can also be extended to estimate the probability of winning games.

In 1982, Maher presented a Poisson regression model to estimate win probabilities based on a team's specific skills (Maher 1982). Poisson scoring models assume goal scoring follows an independent Poisson distribution, and use past performance to obtain maximum likelihood estimates for the number of goals scored by both the home and away team. A benefit of these models is that they can account for additional explanatory variables that affect scoring such as home ice advantage. Another example of a Poisson regression model is the Dixon and Cole model which applies a time weighting model to discount the value of older information (Dixon and Coles 1997). Likewise, Buttrey, Washburn, and Wilson (2011) looked at other explanatory variables by incorporating full strength and non-standard scoring into a Poisson model to predict the goal scoring in games played in the 2009 NHL season (Buttrey et al. 2011).

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Other rating techniques have involved using Google's PageRank algorithm to model pairwise interactions between teams. The PageRank model uses Makov chains to generate ratings by solving the stationary vector of an irreducible stochastic matrix (Brin and Page 1998). When formulating the stochastic matrix, the primary inputs are interactions coded in an adjacency matrix and a vector of additive constants coded in a personalization vector. The model then uses a damping coefficient to adjust how much weight is placed on the adjacency matrix and personalization vector.

Park and Newman were two of the first people to extend the PageRank model to sports rankings and defined a "total win score" to account for a team's direct and indirect wins. This model assumes transitivity of wins to measure indirect wins and controls how much weight is placed on indirect wins by incorporating a damping coefficient (Park and Newman 2005).

In 2008, Govan et al. developed a PageRank model called the Generalized Markov (GeM) that models cumulative margins of victory between pairs of teams with weighted directed edges (Govan, Meyer, and Albright 2008). For this particular model, Govan et al. used a uniform probability vector for the personalization vector and chose an arbitrary value for the damping coefficient. A year later, Govan et al. proposed an "Offense-Defense" model that rated teams based on a combination of distinct offensive and defensive ratings (Govan, Langville, and Meyer 2009).

Then in 2012, Zack et al. revisited Govan's GeM model and experimenting with different ways to model interactions between teams (Zack, Lamb, and Ball 2012). Instead of using margins of victory to weight edges, they generated rankings using four different ways to weight edges (score, total yardage, turnovers, and time of possession). While this research explored different ways to code the adjacency matrix, it also modeled the personalization vector with a uniform probability vector.

Another example of a network-based approaches for sports ranking is Mattingly and Murphy's method of stochastic compartmentation which exploits the natural block structure of a Markov model (Mattingly and Murphy 2010). More recently, Goodman et al. used ten network-based algorithms to rank college football teams and evaluated their performance at predicting weekly outcomes in comparison to the AP Poll rankings (Goodman, Gregory, and Pai 2015).

The primary difference between our model and past applications of the PageRank model is that we incorporate Corsi statistics into a personalization vector, use a nine-fold cross validation to identify tuning parameters, and evaluate out-of-sample prediction accuracy. When the personalization vector is left as a uniform probability vector, it affects all nodes in the same manner and adds no useful information to the model. Therefore, we feel our approach is a useful extension to existing network-based approaches.

The implication of our findings is that adding a personalization and fitting parameter values vector can lead to improved performance when using PageRank models. We found our ratings had a 70% accuracy for predicting the outcome of playoff series over a 9-year period spanning the 2008–2016 seasons and our model outperformed the Colley, Massey, Bradley-Terry, Maher, and GeM models by 5%.

2 Methods

2.1 Calculating the personalized PageRank (PPR) rating

To predict the winner of playoff series we derive ratings from two personalized PageRank (PPR) models. Both models use Corsi statistics to derive a personalization vector, but they differ in how the adjacency matrix is defined. The PPR(wins) model records regular season wins in the adjacency matrix, and the PPR(MoV) model records regular season aggregate margins of victory in the adjacency matrix.

The PageRank of a webpage is often described as the limiting probability of a random web surfer navigating to a particular website while blindly following hyperlinks. However, it is easier to understand the context of our model in terms of a voting scenario. Each team divides its vote among the teams they consider to be the strongest, but the relative influence of each team's vote also depends on how many votes it receives from other teams. By manipulating the personalization vector, we influence the relative importance of a team's vote by modifying the diagonal elements (self-loops) of the adjacency matrix.

2.1.1 Formulation of the adjacency matrix

Mathematically, the adjacency matrix A is a matrix with elements such that

$$A_{ij} = \begin{cases} w_{ij} & \text{if team } i \text{ lost to team } j \\ 0 & \text{otherwise} \end{cases}$$

where w_{ij} is a measure of team j's performance against team i over the entire regular season. Alternately, we can think of an edge as the losing team casting a vote for the winning team.

When using the PPR(wins) model, we measure performance in terms of wins. For example, consider an example where team j plays three games against team i and wins twice. In this example, we would a record a weighted edge of two wins from team i to team j and a weighted edge of one win from team j to team i.

However, some victories are more dominate than others, so we do not have to limit ourselves to modeling interactions between teams based solely on win count. Therefore, we also investigated an approach referred to as the PPR(MoV) model which models aggregate margins of victory between pairs of teams with weighted directed edges.

For example, consider an example where team j wins one game by a score of 4 to 2, loses a second game by a score of 0 to 1, and wins a third game by a score of 2 to 1. Team j would have a margin of victory of 2 goals for the first game and 1 goal for the third game. The two margins of victory would be summed and modeled as a single weighted edge of 3 goals from team i to team j. Likewise, we would record a weighted edge of 1 goal from team j to team i. In many cases this approach provides a reasonable way to record game outcomes between teams, but contextual information can be lost when aggregating margins of victory. For instance, consider game results between teams i and j that are 11-2 and 2-1 in one case, and 5-0 and 5-0 in another. Both cases have a total margin of victory of 10, but the first case involves one blow-out; whereas, the second case has two. Arguably, team i is more dominant in the second example since both games were blow outs.

Despite these limitations, we chose to use the aforementioned methods for two reasons: (1) they are well established in the Colley model which uses win counts as a model input and the Massey and GeM models which use margin of victory and (2) our objective was to illustrate the effects of incorporating a personalization vector into a PageRank model as opposed to presenting additional ways of encoding the adjacency matrix.

2.1.2 Calculation of the personalization vector

To determine an effective statistic to incorporate into our personalization vector, we looked at recent articles focused on the performance of different advanced statistics. A 2015 article featured in the Canadian newspaper, The Globe and Mail, led us to puck possession statistics such as the Corsi and Fenwick statistics (Mirtle 2015). According to this article, teams with better puck possession statistics – even by small margins – generally have more success. MacDonald (2012) also used Corsi statistics to create an expected goals model for evaluating NHL teams and players and notes that Corsi statistics have been shown to perform better than goals for predicting of future goals (MacDonald 2012). While the Corsi statistic is generally regarded as an effective statistic, MacDonald (2012) also notes that other statistics such as the Fenwick statistic have been shown to be more effective than the Corsi (Macdonald, Lennon, and Sturdivant 2012). Likewise, Tulsky et al. (2013) notes that zone entry data is very useful for assessing performance in the offensive or defensive zone (Tulsky et al. 2013).

Advanced statistics can also be adjusted based on varying conditions of play. In 2015, Benjamin Wendorf analyzed the performance of several advanced statistics for various conditions of play (Wendorf 2015). He found that 5-on-5 "close" situations were the strongest metrics for determining series winners. "Close" play refers to games that are tied or within one goal in the first or second periods, or games that are tied in the third period.

We chose to model the personalization vector with a convex combination of the 5-on-5, close Corsi-for and Corsi-against statistics, because we wanted to incorporate offensive and defensive performance measures. We will denote the Corsi-for and Corsi-against statistics for team i as x_i and y_i respectively. The Corsi-for statistic counts the total shots attempted including those that are blocked or miss the net, and it indirectly measures an offense's ability to possess the puck in their offensive zone. The Corsi-against statistic is the same as the Corsi-for statistic except it counts the total shots attempted against a team and indirectly measures a defense's ability to limit puck possession in their defensive zone.

We only consider Corsi statistics for 5-on-5 close play because teams tend to play more defensively when they are leading. This means the trailing team will often take more shot attempts. We use 5-on-5 scenarios because the majority of a game is played at even strength, and because shot attempt data is skewed in favor of the team with a man advantage during powerplays.

To combine these statistics, we first invert the Corsi-against statistic since lower values imply stronger defense. Transforming the Corsi-against statistic in this manner has the desired effect of penalizing teams with high Corsi-against statistics (Figure 1A and B). However, inverting the Corsi-against statistic also induces nonlinearity (Figure 1C and D). Alternatively, a better approach might be to apply a transformation in which we select some constant, C, such

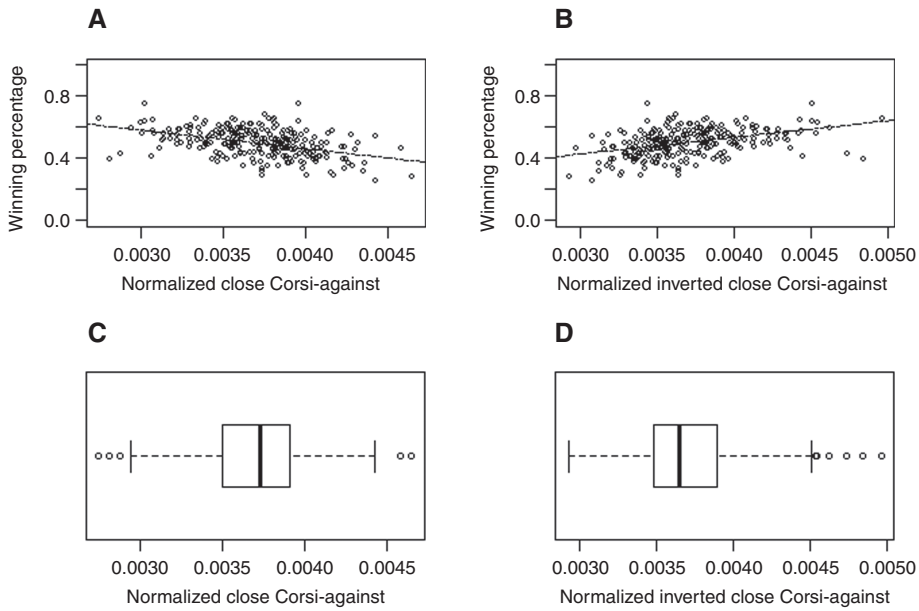

Figure 1: Plots of the (A) raw Corsi-against and (B) inverted close Corsi-against statistics compared to team winning percentages from 2008 to 2016, and box plots of the (C) raw Coris-against and (D) inverted close Corsi-against statistics.

that the difference between C and the raw Corsi-against statistic will always be positive.

Next, we normalize the Corsi-for and inverted Corsi-against statistics so each measure takes on a value from 0 to 1. The normalized Corsi-for, $|x_i|$ and inverted Corsi-against, $|1/y_i|$, statistics for team i are given by dividing by the sum of Corsi-for or inverted Corsi-against statistics for all n teams.

$$|x_i| = x_i \left[\sum_{j=1}^{n} x_j\right]^{-1} \qquad \left|\frac{1}{y_i}\right| = \frac{1}{y_i}\left[\sum_{j=1}^{n} \frac{1}{y_j}\right]^{-1} \tag{1}$$

Then, we combine them using a convex combination where the normalized Corsi-for statistic is weighted with a value of $\beta \in (0, 1)$ and the normalized, inverted Corsi-against statistic is weighted with a value of $(1 - \beta)$.

$$v_i = \beta|x_i| + (1-\beta)\left|\frac{1}{y_i}\right| \tag{2}$$

The last step is to normalize the combined Corsi statistics since the PageRank model requires a stochastic personalization vector.

2.1.3 Formulation of the voting matrix

After defining the adjacency matrix and personalization vector, we create a voting matrix V with elements that represent votes cast by team i for team j. The voting matrix, V, is calculated as

$$V = \alpha DA + (1-\alpha)ev^T \tag{3}$$

where α is a damping coefficient, D is a diagonal matrix with elements $D_{ii} = \left[\sum_{j=1}^{n} A_{ij}\right]^{-1}$, e is a column vector of ones, and v is a personalization vector. The damping coefficient adjusts how much weight is placed on the adjacency matrix and personalization vector.

We then multiply the adjacency matrix A by the diagonal matrix D so that the resulting matrix is row stochastic. This also modifies the matrix so each team allocates its vote proportional to their number of defeats or margins of defeat against all teams. Therefore, teams that are dominated by many others only cast a small proportion of their vote to each of the winning teams. Whereas, teams that rarely lose or lose by a small margin to a few others cast a large proportion of their vote to each of the winning teams. In other words, teams benefit more from winning against quality opponents.

2.1.4 Calculation of the PPR rating

The voting matrix V is an irreducible stochastic matrix. Therefore, a stationary vector π is the dominant eigenvector of V and solves the eigensystem $\pi^T = \pi^T V$. In the context of our model, the stationary vector represents the

proportion of votes cast for team i. The last step for calculating the PPR rating is to scale π by dividing each element by the maximum value contained in π and to multiply the rating by 100. This step is unnecessary in terms of ranking teams, but it makes the rating more intuitive to interpret. When we scale our final ratings in this manner, all teams have a rating between 0 and 100, and the top ranked team has a rating of 100.

2.2 Example PPR calculation for three teams

To illustrate how the PPR(wins) and PPR(MoV) ratings are computed, consider an example where three teams compete against each other over six games. Table 1 shows the outcomes of each game and Table 2 shows the records and 5-on-5 close Corsi statistics for each team.

Figure 2A and B show the network graphs and corresponding adjacency matrices where edges represent aggregate wins and aggregate margins of victory respectively.

Table 1: Game outcomes.

Winner	Loser	Score	Margin of victory
Ants	Bears	4-1	3
Bears	Ants	1-0	1
Ants	Cats	3-1	2
Ants	Cats	2-0	2
Cats	Bears	5-2	3
Bears	Cats	2-1	1

Table 2: 5-on-5 close Corsi statistics.

Team	Record	Corsi-for	Corsi-against
Ants	3-1	55	40
Bears	2-2	42	50
Cats	1-3	42	50

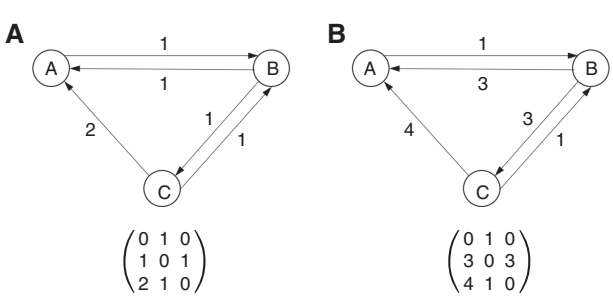

Figure 2: Network graphs and adjacency matrices for (A) wins and (B) margin of victory.

Table 3: PPR(wins) and PPR(MoV) ratings for various values of α and β.

Team	$\beta = 0.25$	$\beta = 0.50$
A. PPR(wins) ratings for $\alpha = 0.25$		
Ants	100	100
Bears	91.5	90.9
Cats	71.9	71.3
B. PPR(wins) ratings for $\alpha = 0.50$		
Ants	97.6	98.0
Bears	100	100
Cats	65.2	65.1
C. PPR(MoV) ratings for $\alpha = 0.25$		
Ants	100	100
Bears	87.7	87.2
Cats	70.2	69.6
D. PPR(MoV) ratings for $\alpha = 0.50$		
Ants	100	100
Bears	96.2	95.9
Cats	63.8	63.5

We then calculate a personalization vector using close Corsi statistics for each team using equation (1), and linearly combine it with the adjacency matrix to form the voting matrix in equation (2). Last, we iterate $\pi^T = \pi^T V$ until π^T converges to a stationary vector and scale the ratings so all teams have a rating between 0 and 100.

Selecting effective parameter values for α and β is critical because the ratings, and subsequent rankings, are sensitive to the model inputs. For instance, setting the damping coefficient to 0.25 results in the Ants having the highest rating when using the PPR(wins) model (Table 3A). However, if the damping coefficient is set to 0.5, the Bears are rated as the top team (Table 3B).

In contrast, the ratings are less sensitive to changes made to the β parameter. In this example, changing the β results in a trivial adjustment to the ratings and does not alter the rankings.

3 Results

In this section, we present optimal tuning parameters identified through a nine-fold cross validation grid search and report the prediction accuracy of our model when using the optimal parameter values. The cross validation process involved using a grid search to identify tuning parameters for training sets consisting of eight seasons (120 playoff series). We repeated this process nine times by leaving out a single season (15 playoff series) in each iteration. Using this approach allowed us to evaluate the

stability of the tuning parameters with respect to changing the training data.

We then compared our final model's prediction accuracy with predictions based on raw Corsi statistics and the Colley, Massey, Bradley-Terry, Maher, and GeM rating models. Last, we provide sensitivity analysis to assess the robustness of our model's performance to changes in parameter values.

Games logs were obtained from http://www.hockey-reference.com/ and Corsi statistics from http://puckalytics.com/teamstats.html which contains Corsi statistics by team for the 2008–2016 seasons. Since the Corsi statistics were not available for seasons prior to 2008, we were only able to extend our analysis back to the 2008 season.

3.1 Cross validation results and parameter selection

To identify tuning parameters, we employed a grid search allowing us to conduct an exhaustive parameter search over the parameter space (Hsu, Chang, and Lin 2003). There are other, more advanced, methods for identifying the parameter values, but this simple method allowed us to evaluate all parameter values with little computational effort.

Figure 3 shows an example density map for the PPR(wins) model for the 2009–2016 training set. Darker colors represent lower prediction accuracy and the locations of the best-fit parameter values are marked with an "X".

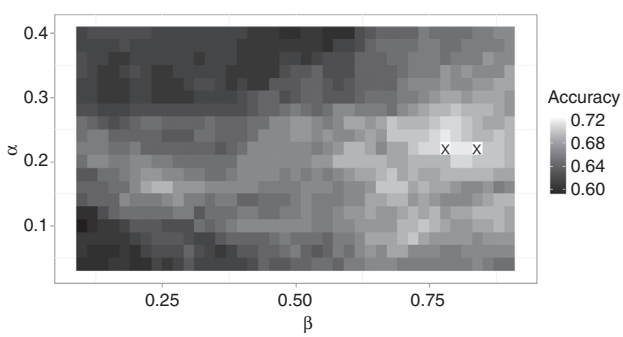

Figure 3: Example PPR(wins) density map for a training set consisting of the 2009–2016 seasons.

We found the tuning parameters for the PPR(wins) model were stable across the folds since all of the grid searches identified 0.22 as the best-fit value for the α parameter, and 8 of the 9 grid searches identified 0.78 as a best-fit value for the β parameter (Table 4). Since the Corsi-against statistic was consistently weighted more heavily than the Corsi-for statistic, our results suggest limiting an opponent's shot attempts may be more important in post-season play than generating shot attempts.

In contrast, the PPR(MoV) tuning parameters were fairly stable for the α parameter but unstable for the β parameter. We found that 6 of the 9 grid searches identified 0.14 as a best-fit value for the α parameter and 4 of 9 grid searches identified 0.16 as a best-fit value (Table 5). However, the best-fit β parameters ranged from 0.3 to 0.9 in value.

3.2 Comparison of PPR models with existing models

The PPR(wins) model had the best performance with an average accuracy of 70%, and the PPR(MoV) model had an average accuracy of 62% (Table 6). In contrast, the team with a higher regular season point total won only 53% of the games. The best years for the PPR(wins) rating were 2010, 2011, 2013, 2015, and 2016 at 80%, and the worst years was 2012 at 40%. In comparison, the best years for a points-based prediction were 2011, 2009, and 2008 at 67%, and the worst year was 2012 at 33%. The raw Corsi-for and Corsi-against statistics performed relatively well with average accuracies of 62% and 59% respectively.

The second best rating model in terms of overall accuracy were the Massey and Maher models, which had average accuracies of 65%. Both models outperformed the PPR(wins) model in the 2008 season, and the Maher model outperformed the PPR(wins) model in the 2009 season.

3.3 Sensitivity analysis

In this section, we discuss the sensitivity of our PPR(wins) results to the selection of parameter values. We compare

Table 4: PPR(wins) cross-validation results.

Parameters (α, β)	Test season									Average
	2008	2009	2010	2011	2012	2013	2014	2015	2016	
(0.22, 0.78)	X	X	X	X	X		X	X	X	
(0.22, 0.84)	X		X	X	X	X	X		X	
Prediction accuracy	0.53	0.60	0.80	0.80	0.40	0.80	0.73	0.80	0.80	0.70

An "X" identifies the tuning parameters that were used to calculate the out-of-sample prediction accuracies.

Table 5: PPR(MoV) cross-validation results.

Parameters (α, β)	Test season									Average
	2008	2009	2010	2011	2012	2013	2014	2015	2016	
(0.08, 0.68)	X									
(0.14, 0.30)	X	X	X	X				X		
(0.14, 0.34)	X	X						X		
(0.14, 0.74)					X					
(0.14, 0.90)					X					
(0.16, 0.34)						X			X	
(0.16, 0.76)	X				X					
(0.16, 0.88)					X					
(0.18, 0.90)					X					
(0.34, 0.36)						X	X			
(0.38, 0.86)					X					
Prediction accuracy	0.53	0.60	0.66	0.80	0.37	0.77	0.60	0.57	0.66	0.62

An "X" identifies the tuning parameters that were used to calculate the out-of-sample prediction accuracies.

Table 6: Comparison of prediction accuracy by rating method.

Rating method	2008	2009	2010	2011	2012	2013	2014	2015	2016	Average
PPR(wins) model	0.53	0.60	0.80	**0.80**	0.40	**0.80**	**0.73**	**0.80**	**0.80**	**0.70**
Maher model	**0.93**	**0.66**	0.60	0.73	0.33	0.66	0.66	0.73	0.60	0.65
Massey model	0.86	0.60	0.73	0.73	0.40	0.73	0.46	0.66	0.66	0.65
PPR(MoV) model	0.53	0.60	0.66	**0.80**	0.37	0.77	0.60	0.57	0.66	0.62
Corsi-for statistic	0.53	0.60	**0.86**	0.66	0.33	0.66	0.60	0.60	0.73	0.62
GeM model	**0.93**	**0.66**	0.53	**0.80**	0.33	0.53	0.53	0.66	0.46	0.61
Corsi-against statistic	0.53	0.60	0.53	0.60	**0.53**	0.66	0.66	0.53	0.60	0.59
Colley model	0.66	0.53	0.66	0.60	0.40	0.60	0.53	0.60	0.60	0.57
Bradley-Terry model	0.73	0.66	0.66	0.66	0.33	0.60	0.40	0.53	0.60	0.57
Regular season points	0.66	0.60	0.46	0.66	0.33	0.60	0.40	0.46	0.53	0.53

Bold values indicate the highest prediction accuracy for each season.

the out-of-sample accuracies based on parameter values that spanned the entire parameter space, values that were reasonably close to the tuning parameters, and a point estimate for the tuning parameters. Here, we arbitrarily define "reasonably close" as values within 0.1 of the tuning parameter point estimate. Overall, we found that using values reasonably close to the tuning parameter point estimate improved for the PPR(wins) model performance for six of the nine seasons.

Due to the instability of the PPR(MoV) tuning parameters, we were unable to justify a suitable estimate for the tuning parameters to apply to the entire data set. Therefore, we excluded the PPR(MoV) model from this section.

If we made no attempt to train our model on past seasons and randomly selected α and β parameter values ranging from 0 to 1, the PPR(wins) model had an average prediction accuracy of 60.3% with a standard deviation of 8.4% (Table 7). However, if we chose parameter values that were reasonably close to the most common tuning parameters identified in Section 3.1 ($\alpha = 0.22$, $\beta = 0.78$), the average prediction accuracy increased to 66.9% with a standard deviation of 4.3%. These results show that restricting the parameter values to areas reasonably close

Table 7: Sensitivity of PPR(wins) average out-of-sample accuracy to parameter selection.

Season	Parameter selection method		
	Random	Best-fit area estimate	Best-fit point estimate
	$\alpha \in (0, 1)$ $\beta \in (0, 1)$	$\alpha = 0.14 \pm 0.1$ $\beta = 0.78 \pm 0.1$	$\alpha = 0.14$ $\beta = 0.78$
2008	**0.560 ± 0.090**	0.526 ± 0.022	0.530
2009	**0.620 ± 0.049**	0.601 ± 0.034	0.600
2010	0.607 ± 0.095	**0.754 ± 0.050**	0.800
2011	0.614 ± 0.071	**0.734 ± 0.041**	0.800
2012	**0.430 ± 0.072**	0.404 ± 0.037	0.400
2013	0.684 ± 0.109	**0.798 ± 0.044**	0.800
2014	0.593 ± 0.109	**0.699 ± 0.039**	0.730
2015	0.691 ± 0.089	**0.737 ± 0.071**	0.800
2016	0.626 ± 0.076	**0.765 ± 0.052**	0.800
Average	0.603 ± 0.084	**0.669 ± 0.043**	0.696

Bold values indicate the highest prediction accuracy for each season.

to the tuning parameters increased the average prediction accuracy and reduced variance. Therefore, we concluded the model fitting process was generally effective, and our results were fairly robust to small changes in the tuning parameters.

In terms of individual season performances, the average prediction accuracy increased for six of the nine seasons when using parameters reasonably close to the tuning parameters. In the seasons with a lower prediction accuracy (2008, 2009, and 2012), the largest drop in accuracy was a 3.4% drop for the 2008 season. Furthermore, the PPR(wins) poor performance at predicting the 2012 season might be attributed to the fact that the teams only played 48 regular season games due to a lock out.

Since the results in Table 7 suggest the prediction outcomes take on continuous values, we provided bubble charts to capture the discrete nature of our prediction results. Figure 4 conveys a visual representation of the distribution of out-of-sample accuracies where point size corresponds to the relative frequency of each prediction outcome.

4 Conclusion

In this study, we have demonstrated an effective rating system for predicting the NHL playoffs. Similar to previous research, we adapted the PageRank model to fit the context of hockey by creating a network that models interactions between teams. However, unlike previous research we also incorporated Corsi statistics into a personalization vector, used a nine-fold cross validation to identify tuning parameters, and evaluated out-of-sample prediction accuracy.

Our PPR(wins) model performed well with an average out-of-sample prediction accuracy of 70% and outperformed alternative rating models by 5% over nine seasons. Additionally, the PPR(wins) model had stable tuning parameters, and the achieved a prediction accuracy of 67% when using parameters reasonably close to the tuning parameters. Therefore, we concluded our results were fairly robust to minor changes to the tuning parameters. In contrast, the PPR(MoV) method was far less effective with an out-of-sample prediction accuracy of 62% and unstable tuning parameters.

In summary, our results suggest that fitting parameter values and adding a personalization vector can lead to improved performance when using PageRank models. We offer two examples of how team statistics can be incorporated into the personalization vector of the PageRank model. However, there are countless ways to modify the PageRank model. Our approach can be extended to incorporate alternative statistics into the personalization vector and different methods for coding the adjacency matrix. For instance, we could record goal differentials in the personalization vector and wins in the adjacency matrix or head-to-head Corsi statistics into the adjacency matrix and wins in the personalization vector. Additionally, data found on

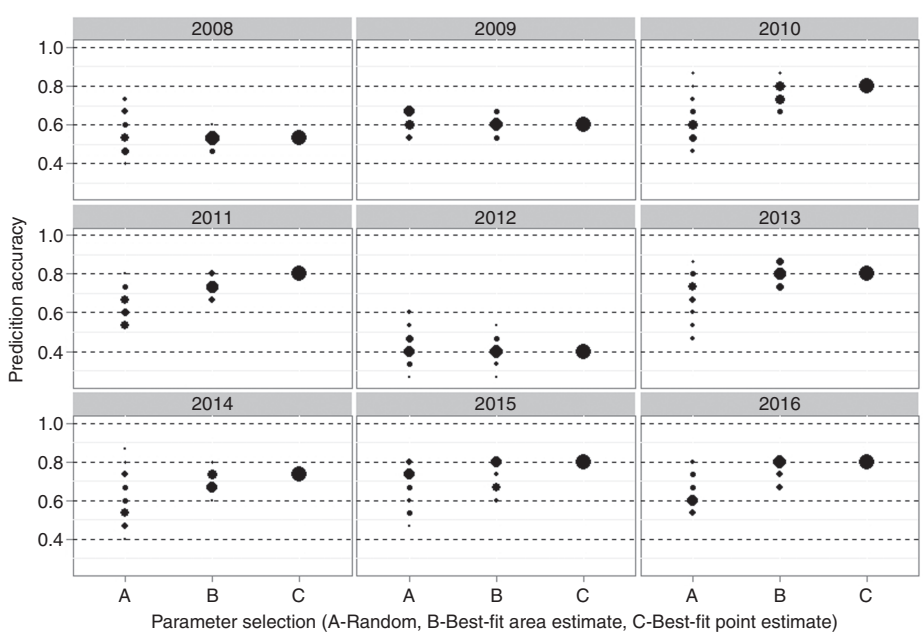

Figure 4: Sensitivity of PPR(wins) out-of-sample accuracy to parameter selection.

http://www.corsica.hockey/data/ could be used to extend our approach to game by game analysis as opposed to just post-season play. Numerous improvements or updates could be made to the presented model, but we believe the use of the personalization vector is the key to seeing the potential of the Page-Rank model.

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